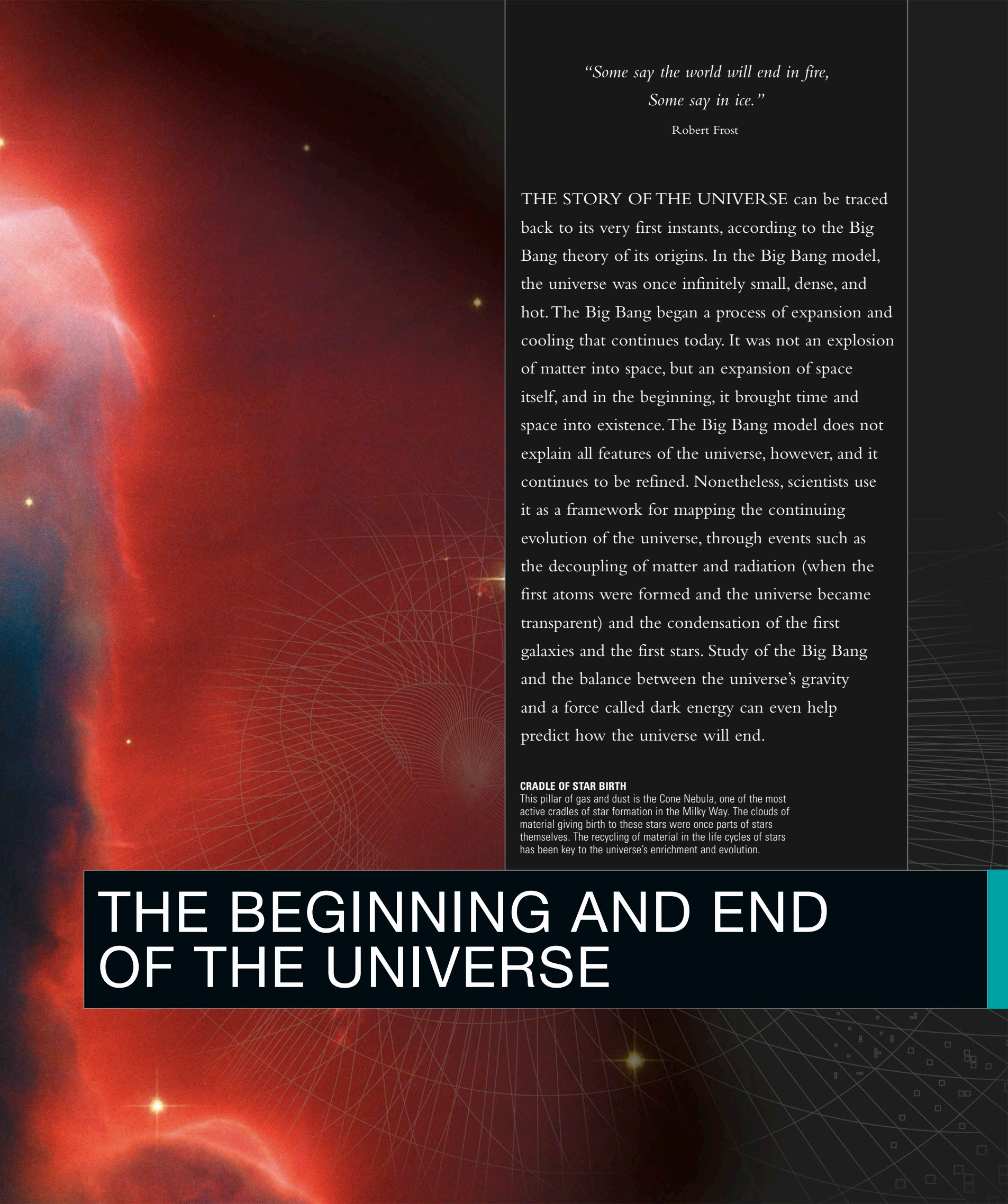




*"Some say the world will end in fire,
Some say in ice."*

Robert Frost

THE STORY OF THE UNIVERSE can be traced back to its very first instants, according to the Big Bang theory of its origins. In the Big Bang model, the universe was once infinitely small, dense, and hot. The Big Bang began a process of expansion and cooling that continues today. It was not an explosion of matter into space, but an expansion of space itself, and in the beginning, it brought time and space into existence. The Big Bang model does not explain all features of the universe, however, and it continues to be refined. Nonetheless, scientists use it as a framework for mapping the continuing evolution of the universe, through events such as the decoupling of matter and radiation (when the first atoms were formed and the universe became transparent) and the condensation of the first galaxies and the first stars. Study of the Big Bang and the balance between the universe's gravity and a force called dark energy can even help predict how the universe will end.

CRADLE OF STAR BIRTH

This pillar of gas and dust is the Cone Nebula, one of the most active cradles of star formation in the Milky Way. The clouds of material giving birth to these stars were once parts of stars themselves. The recycling of material in the life cycles of stars has been key to the universe's enrichment and evolution.

THE BEGINNING AND END OF THE UNIVERSE